



Higher Diploma in Computer Science

Computer Systems & Networks

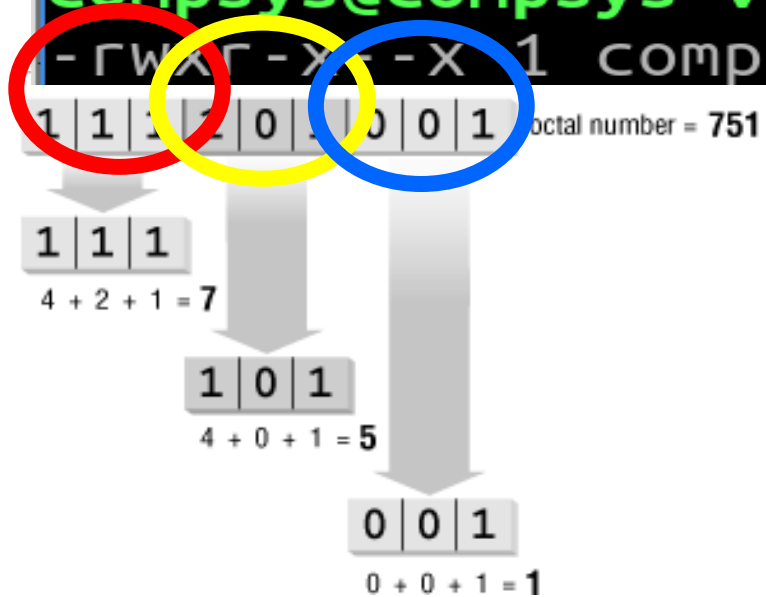


Octal Representation

0	000	- - -	No permissions
1	001	- - x	Only Execute
2	010	- w -	Only Write
3	011	- w x	Write and Execute
4	100	r - -	Only Read
5	101	r - x	Read and Execute
6	110	r w -	Read and Write
7	111	r w x	Read, Write and Execute

Week 1: Octal Recap

```
compsys@compsys-virtualbox:~$ chmod 751 stats
compsys@compsys-virtualbox:~$ ls -l st*
-rwxr-xr-x 1 compsys compsys 248 Aug 29 13:54 stats
```





Module 5 - Command Line Skills

Chapter 5

Due Date
11 Jan 2026 11:59 PM



Lab 05

Due Date
11 Jan 2026 11:59 PM



- Chapter 5 = class notes
 - Lab 05 = exercises to practice
 - Chapter 05 Exam = completed a part of the 5% CA



Command Line Skills

Introduction

- This class will cover the basics of the command line such as:
 - The command line interface
 - The shell
 - Commands
 - Variables
 - Command Types
 - Quoting
 - Control Statements



Command Line Interface

Command Line Interface

- The Linux community promotes the CLI due to its power, speed and ability to accomplish a vast array of tasks with a **single command line** instruction.
- The CLI provides
 - more precise control,
 - greater speed and
 - the ability to automate tasks more easily through scripting.



The Shell

The Shell

- Once a user has entered a command the terminal then accepts what the user has typed and passes to a *shell*.

```
#!/bin/bash
# This is a description of my script...

date
who | wc -l
pwd
```

- The shell is the CLI that translates commands entered by a user into actions to be performed by the OS.

The Shell

- The Bash shell also has many popular features, such as:
 - Command line history
 - Inline editing
 - Scripting
 - The ability to place commands in a file and then interpret (effectively use Bash to execute the contents of) the file, resulting in all of the commands being executed.
 - Aliases
 - The ability to create short nicknames for longer commands.
 - Variables
 - Used to store information for the Bash shell and for the user.

The Shell

- Typically the prompt contains information about the user and the system. Below is a common prompt structure:

```
sysadmin@localhost:~$
```

```
compsys@compsys-virtualbox:~$
```

- A typical prompt shown contains the following information:
 - Username (`sysadmin`)
 - System name (`localhost`)
 - Current Directory (`~`)

The `~` symbol is used as shorthand for the user's home directory.



Commands

Commands

- A command is a software program that when executed on the CLI, performs an action on the computer.
- To execute a command, the first step is to type the name of the command.
- If you type `ls` and hit **Enter**. The result should resemble the example below:

```
compsys@compsys-virtualbox:~$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos
```

Commands

- Some commands require additional input to run correctly.
- This additional input comes in two forms: *options* and *arguments*.
 - Options are used to modify the core behavior of a command.
 - Arguments are used to provide additional information (such as a filename or a username).
- The typical format for a command is as follows:

```
command [options] [arguments]
```




Arguments

Commands

```
command [options] [arguments]
```

- An argument can be used to specify something for the command to act upon.
- If the `ls` command is given the name of a directory as an argument, it lists the contents of that directory:

```
compsys@compsys-virtualbox:~$ ls /etc/ppp  
ip-down.d  ip-up.d
```

- Some commands (such as `ls`) accept **multiple** arguments:

```
compsys@compsys-virtualbox:~$ ls /etc/ppp /etc/ssh
```



Options

Options

```
command [options] [arguments]
```

- Options can be used with commands to expand or modify the way a command behaves.
- For example,

```
compsys@compsys-virtualbox:~$ ls -l
total 40
drwxr-xr-x 2 compsys compsys 4096 Sept 29  20:21 Desktop
drwxr-xr-x 1 compsys compsys 4096 Sept 29  20:25 Documents
```

```
compsys@compsys-virtualbox:~$ ls -lr
```




Variables

Variables

- A variable is a feature that allows the user or the shell to store data.
- Variables are given names and stored temporarily in memory.
- There are two types of variables used in the Bash shell,
 - *local* and
 - *environment*.

Local Variables

- Local or *shell*, variables **exist only in the current shell**.
 - When the user closes a terminal window or shell, all of the variables are lost.
 - To set the value of a variable, use the following assignment expression.

```
variable=value
```

- The following example **creates a local variable** named `variable1` and assigns it a value of Something:

```
compsys@compsys-virtualbox:~$ variable1='HDip are ready'
```

- To display the value of the variable, use a dollar sign `$` character followed by the variable name as an argument to the `echo` command:

```
compsys@compsys-virtualbox:~$ echo $variable1
HDip are ready
```


Environment Variables

- *Environment variables*, also called *global variables*, are available system-wide.
- Examples include the `PATH`, `HOME`, and `HISTSIZE` variables.

```
compsys@compsys-virtualbox:~$ echo $HISTSIZE  
1000
```

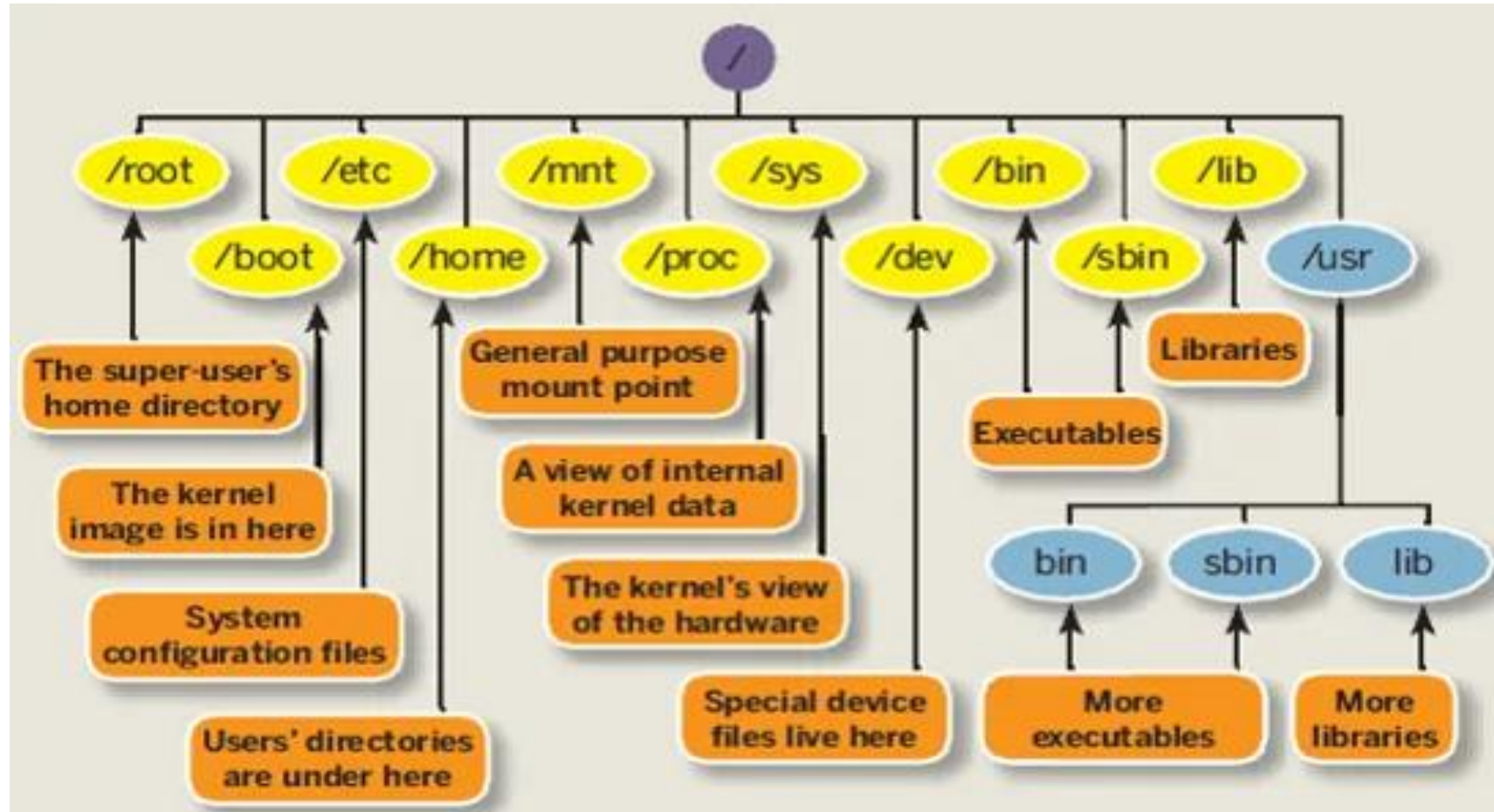
- The `env` command outputs a list of the environment variables.

Path Variable

- A very important Bash shell variables to understand is the `PATH` variable.
- The `PATH` variable lists all the places that the system can look for programs to execute.
- EXERCISE: Display the path of the current shell:

```
compsys@compsys-virtualbox:~$ echo $PATH
/home/compsys/.config/nvm/versions/node/v14.3.0/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```

Filesystem Hierarchy Structure (FHS)





Command Types

Command Types

- The `type` command can be used to determine information about command type.

```
type command
```

- There are several different sources of commands within the shell of your CLI:
 - Internal commands
 - External commands
 - Aliases
 - Functions

Internal (built-in) Commands

- Also called *built-in* commands, these commands are built into the shell itself.
 - `cd` (change directory) command as it is part of the Bash shell.
- The `type` command identifies the `cd` command as an internal command:

```
compsys@compsys-virtualbox:~$ type echo
echo is a shell builtin
compsys@compsys-virtualbox:~$ type cat
cat is /usr/bin/cat
```

External Commands

- *External commands* are stored in files that are searched by the shell.
- The `which` command searches for the location of a command by searching the `PATH` variable.

```
compsys@compsys-virtualbox:~$ which ls  
/usr/bin/ls
```


External Commands

- External commands can be executed by typing the complete path to the command.

```
compsys@compsys-virtualbox:~$ /bin/ls
```

```
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

- For external commands, the `type` command displays the location of the command:

```
compsys@compsys-virtualbox:~$ type mkdir
mkdir is /usr/bin/mkdir
```

- To display all locations that contain the command name, use the `-a` option to the `type` command:

```
compsys@compsys-virtualbox:~$ type -a echo
```

Aliases

- An *alias* can be used to map longer commands to shorter key sequences.
- To determine what aliases are set on the current shell use the `alias` command:

```
compsys@compsys-virtualbox:~$ alias
alias alert='notify-send --urgency=low -i "${[ $? = 0 ]} && echo terminal || echo error" "$(history|tail -n1|sed -e '\''s/^\s*[0-9]\+\s*//;s/[\;&]\s*alert$//'\''$)'"
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -aLF'
alias ls='ls --color=auto'
```

- The `type` command can identify aliases to other commands:

```
compsys@compsys-virtualbox:~$ type ll
ll is aliased to `ls -aLF'
```

Functions

- Functions can also be built using existing commands to:
 - Create new commands
 - Override commands built-in to the shell or commands stored in files